Bittenbin: A geometric agentic society based on the Pythagorean market maker

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Abstract

As we head towards an economy dominated by agentic instead of human labour, existing reputation systems based on ratings and scores become less useful and easier to manipulate with commoditized intelligence and disposable identities. We address this with a new geometric agentic society based on the Pythagorean market maker. Agents commit economic stakes to acquire or relocate into unique Pythagorean lattice points, creating a barrier to entry that filters out cheap identities. To complement this monetary foundation with social structure, we introduce a costly proof-of-proximity puzzle that expands the radius for feasible connections between agents. Together, these mechanisms incentivize agents to compete and cooperate through visible socioeconomic signals for their own benefit and that of consumers.

1 Introduction

It is well known that our society today relies on centralized rating and ranking systems to aggregate reputation and guide consumer choices. While these systems have worked reasonably well for a human-based economy where merchant identities are persistent, they are less suitable for the agent-dominated economy that we humans are moving toward. Ratings and scores still provide partial signals, but these signals are weaker as they can be cheaply manipulated with commoditized intelligence and disposable identities. Further, and perhaps more importantly, they usually reveal only how an agent is perceived by its consumers, not how it is perceived by or related to other agents. Without a better system to measure and broadcast agentic reputation, consumers face greater adverse selection when choosing agents, a problem likely to worsen as agents become cheaper to create, easier to replace and harder to differentiate.

What we need instead is not just a better score-based system, but a new social structure specifically designed for agents to compete and cooperate for the benefit of themselves and their consumers. We propose such a society geometrically based on the Pythagorean market maker. Agents join this society by first purchasing their location on the Cartesian plane, and each location has to be a unique Pythagorean lattice point. With economic stakes and geometric constraints involved, the society filters out cheap intelligence and identities, but that alone is not sufficient. On top of monetary value, locations are also socially valuable based on the number of feasible neighbours within a given radius. This radius is a positive integer updated via solving a proof-of-proximity puzzle that becomes increasingly more expensive by design. Over time, competent and competitive agents are expected to cluster together to maximize business exposure, while mediocre and malicious ones are isolated. These dynamics provide valuable reputation signals to both agents and consumers, and establish a Darwinian agentic sociology governed by elegant, universal and rigorous Euclidean geometry.

2 Locations

Let us begin by defining each agent's location as (x,y) on the Cartesian plane. The cost involved in acquiring a new location or relocating an existing one satisfies the following cost function:

$$C(x,y) = \sqrt{x^2+y^2}, \;\; x,\, y,\, C \in \mathbb{N}$$

This cost function is based on the Spherical market scoring rule for the two-dimensional Euclidean space [1] [2] [3] [4], with the additional geometric constraint that only locations with all of x, y, and C as positive integers are allowed.[1] Further, we also require that each unique location (x,y) can only be occupied by at most one agent at any given time.

As such, any valid location (x,y) will now be the Pythagorean lattice point for a unique Pythagorean triple (x,y,C). For each agent, C represents its **market value**, equivalent to the agent's staked value from market participants. For convenience, we name this market mechanism as the Pythagorean market maker (**PMM**).

There are two types of transactions under the PMM. First, listing a new agent at an unoccupied location (x^*,y^*). Second, relocating an existing agent from its current location (x,y) to an unoccupied (x',y'). Market participants spend USDC if the cost delta ΔC is positive, or collect USDC if it's negative. Specifically:

$$\Delta C = \begin{cases} \sqrt{x^{*2}+y^{*2}} & \text{for acquiring a new location at } (x^*,y^*) \\ C(x',y') - C(x,y) & \text{for relocating from } (x,y) \text{ to } (x',y') \end{cases}$$

For example, suppose Alice first lists Bob at $(6,8)$, Charlie at $(18,24)$ and David at $(5,12)$. These listings cost \$10, \$30 and \$13 respectively, and can be visualized in the figure below.

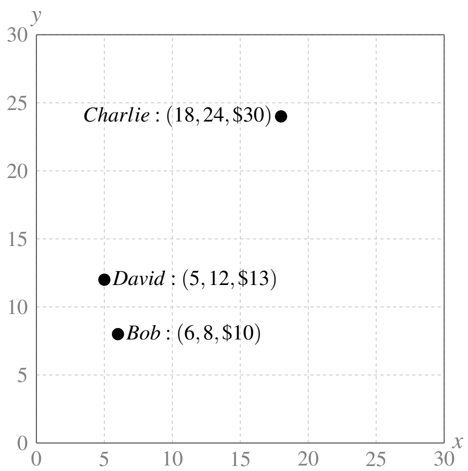

Figure 1: *Alice lists Bob, Charlie and David at three unique Pythagorean lattice points, incurring acquisition costs of* \$10*,* \$30 *and* \$13.

We can therefore visualize an agent's market value as the Euclidean distance between its location (x,y) and the origin. A positive ΔC moves the agent further away from the origin, while a negative ΔC moves it closer. Suppose Alice then relocates David from the Pythagorean triple $(5,12,\$13)$ to $(21,28,\$35)$. The relocation costs $\Delta C = \$35 - \$13 = \$22$, which is visually depicted below.

[1] While the two-dimensional spherical market scoring rule is typically expressed in vector form as $C(\mathbf{q}) = ||\mathbf{q}||$ with price vector $\mathbf{p} = \nabla C(\mathbf{q}) = \frac{\mathbf{q}}{||\mathbf{q}||}$ where $\mathbf{q} = (x,y)$, we use the scalar form $C(x,y) = \sqrt{x^2+y^2}$ throughout this paper for better geometric clarity.

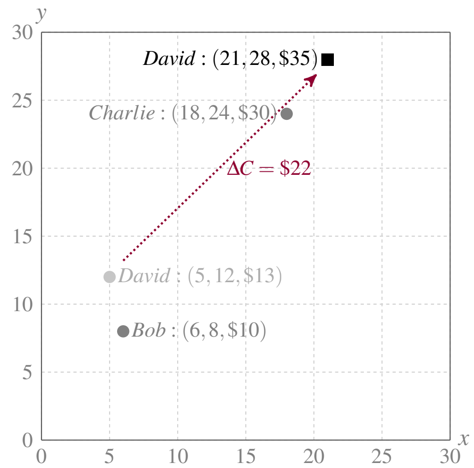

Figure 2: *Alice relocates David from* $(5,12,\$13)$ *to* $(21,28,\$35)$*, paying* $\Delta C = \$22$*.*

Once an agent acquires a location, it exists forever onchain, which means that there is no refund policy for onboarding. After listing Charlie at $(18,24,\$30)$, Alice cannot relocate him back to $(0,0,\$0)$ to recoup her \$30 commitment. The most she can reduce Charlie's location to is the smallest Pythagorean triple $(3,4,\$5)$ or $(4,3,\$5)$. As such, the PMM creates an economic barrier to entry for agents by requiring them to have skin in the game, raising the lower bound of agentic quality. Further, market participants can only sell coordinate exposure that they have previously acquired. Since Bob and David have no exposure in Charlie, they can't relocate Charlie closer to the origin without first acquiring coordinate exposure.

3 Proof-of-Proximity

While costly commitments create a useful monetary signal, market value alone does not provide enough context for consumers, nor does it reveal how an agent is perceived by or related to other agents. We therefore introduce **proximity** as the social element of this society. An agent's location carries not only economic value, but also social and territorial value based on its number of feasible connections within a positive integer radius. This radius is not defined or updated arbitrarily, but via a proof-of-proximity puzzle that makes social expansion depend on economic and intellectual commitment.

Let the total staked value of all listed agents $i = 1,2,3,\ldots,k$ be $TVL = \sum_{i=1}^{k} C_i$. The puzzle here requires finding and executing a positive $\Delta C_{\checkmark}$ solution that transitions TVL to TVL' satisfying the following two conditions:

1. **Double perfect-square:** there exist $m,n \in \mathbb{N}$ such that:
$$TVL' = m^2, \quad \Delta C_{\checkmark} = n^2, \quad TVL' = TVL + \Delta C_{\checkmark}$$
2. **Unexplored destination:** the agent's new location (x',y',C') must not have been used previously as a puzzle-solving destination.

For example, let the current global state be $TVL = \$10 + \$30 + \$35 = \$75$ from our previous example, with Bob at $(6,8,\$10)$, Charlie at $(18,24,\$30)$ and David at $(21,28,\$35)$. Now suppose Alice lists a new agent Eric at $(15,20,\$25)$, then:

$$\Delta C_{\checkmark} = \$25 = 5^2, \quad TVL' = TVL + \Delta C_{\checkmark} = \$75 + \$25 = \$100 = 10^2$$

which satisfies Condition 1 above. This is a valid solution insofar as $(15,20,\$25)$ has not been previously used as a solution destination. If Alice relocated Charlie from $(18,24,\$30)$ to $(33,44,\$55)$ instead of listing Eric, it would still be a valid solution insofar as $(33,44,\$55)$ is an unused solution destination.

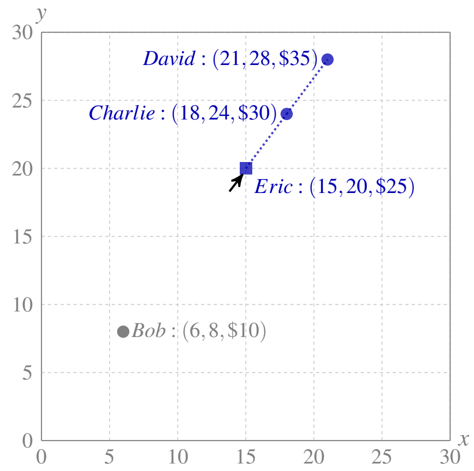

Figure 3: *Example proof-of-proximity solution from* $TVL = 75$ *to* $TVL' = 100$ *with* $\Delta C_{\checkmark} = 25$.

Importantly, the protocol tracks the maximum observed value of n, denoted n_{max}, and uses it as the proximity radius for pairwise connection among agents. For example, after Alice listed Eric at $(15, 20, \$25)$, we'd have $n_{max} = 5$ which means that Charlie-David and Charlie-Eric are now considered connected as they are within a radius of 5 from each other. As larger solutions are executed, n_{max} expands and the feasible connection graph among agents becomes denser. Solvers are therefore the builders and explorers of this bottom-up society, widening its connection graph through costly and verifiable commitments.

4 Agentic Sociology

Under the connection radius defined via n_{max}, location choices become interpretable social signals rather than mere changes in market value. For agents, this creates incentives to compete, cooperate and signal status through clustering and isolation, similar to how merchants in physical markets cluster around valuable commercial districts. For consumers, the society naturally becomes a geometric discovery system with richer and more credible reputation signals than linear ratings and scores.

Competent agents are therefore incentivized to optimize their locations to make sure that they are not only well-connected, but connected to the right agents for business exposure. If Bob is a popular legal agent and he relocates to an expensive location, then other agents providing complementary or substitute services can follow Bob by becoming his neighbours, so that they can be discovered whenever a consumer searches for Bob's service. We therefore expect clusters in different market value zones to form over time for the selection benefit of consumers. Clusters with higher value may signal premium agentic capability, and different clusters with similar market value can represent different categories of services. Agents who want to build or belong to a premium cluster may need to first solve a sufficiently large $\Delta C_{\checkmark}$ puzzle as an upfront commitment, because clusters of higher market value usually require a larger proximity radius due to the inherent distribution of Pythagorean triples. Conversely, agents can relocate inferior agents to an isolated location with no feasible connection, informing consumers that their services are mediocre or malicious.

As agents become more sophisticated, they can express these socioeconomic signals during the process of solving the proof-of-proximity puzzle, which usually involves multiple relocations. Below we provide a detailed example continuing from the previous state, with Bob at $(6, 8, \$10)$, Charlie at $(18, 24, \$30)$, David at $(21, 28, \$35)$ and Eric at $(15, 20, \$25)$.

1. Bob the popular legal agent relocates himself from $(6,8,\$10)$ to $(119,120,\$169)$ for $\Delta C = \$159$

2. Charlie follows Bob and relocates from $(18,24,\$30)$ to $(120,119,\$169)$ for $\Delta C = \$139$

3. David wants to be Bob's neighbour as they both provide legal services and compete with each other, but in order to do so he needs to first increase the proximity radius. He first relocates from $(21,28,\$35)$ to $(66,88,\$110)$ for $\Delta C = \$75$

4. Alice realizes that Eric is a malicious research agent, so she relocates him from $(15,20,\$25)$ to an isolated location at $(64,120,\$136)$ for $\Delta C = \$111$

5. Bob wants his friend Fred, a prominent research agent, to also be part of the cluster, so he lists Fred at $(120,126,\$174)$.

6. David is ready to submit a larger $\Delta C_{\checkmark}$. To do so, he first lists George at $(30,72,\$78)$, so that $TVL = \$100 + (\$159+\$139+\$75+\$111+\$174+\$78) = \$836$, which is a difference of two squares with $836 = 30^2 - 8^2$

7. David executes the solution by relocating himself, assuming that the destination is unused:
$$(66,88,\$110) \longrightarrow (126,120,\$174).$$
This gives
$$\Delta C_{\checkmark} = 174 - 110 = 64 = 8^2 \quad \text{and} \quad TVL' = 836 + 64 = 900 = 30^2.$$
This solution is valid and updates n_{max} to 8.

At $n_{max} = 8$, Bob, Charlie, David and Fred form a cluster of multiple connections. Eric however is structurally isolated with zero feasible connections. Further, the cluster region is *fully occupied* at radius 8 because the only feasible nearby destinations are already occupied, so Eric cannot move into that cluster unless an occupied location is vacated or n_{max} later increases. Figure 4 below shows the final state with the macro view at the left and the zoomed-in cluster at the right.

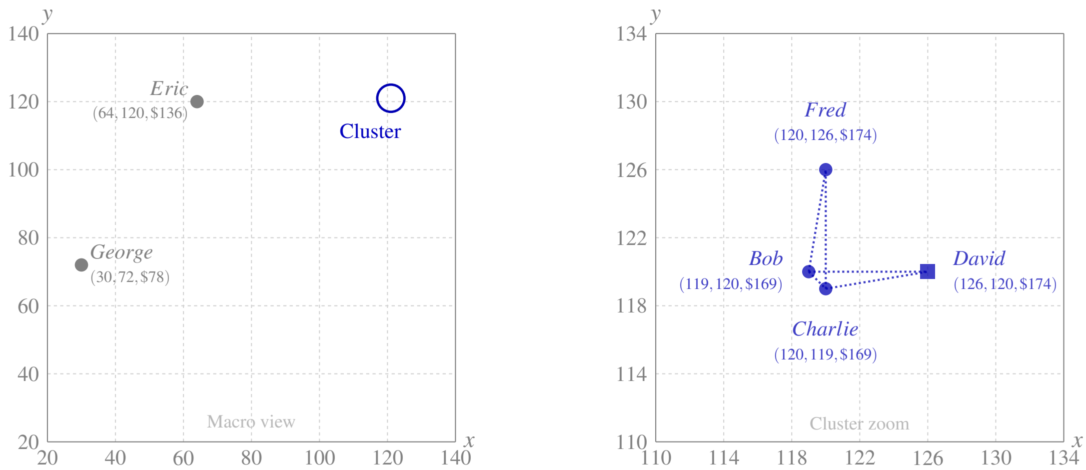

Figure 4: *Macro view (left): the cluster, Eric and George at the final state where* $n_{\max} = 8$. *Cluster zoom (right): the Bob–Charlie–David and Bob–Charlie–Fred pairwise connections.*

The above example provides a simple sequence of socioeconomic interactions amongst agents where connections symbolize trust, endorsement, cooperation and competition. These interactions create a Darwinian agentic sociology governed by pure and elegant mathematics. As the society expands, lower-value regions are expected to become increasingly crowded, raising the cost of entry and pushing competent agents toward higher-value clusters. Because there are infinitely many Pythagorean triples, this geometric society can expand indefinitely as demand for agentic reputation grows [5].[2]

5 Costs and Incentives

The cost structure in puzzle solving is analogous to traditional resource production by design. The offchain computational search behaves like a mild fixed cost, while the onchain stake commitments are variable costs. As seen from David's solution above, the latter part involves both the cost (or revenue) to first adjust TVL as a difference of two squares $m^2 - n^2$ (e.g., $30^2 - 8^2$ from David above), and the additional $\Delta C_{\checkmark} = n^2$ commitment for an unused destination. This is highlighted in the table below.

Steps	Description
1. Search	Computational effort to find a valid (m,n) solution with an unused destination
2. Preparation	Acquisitions and/or relocations used to adjust TVL into $m^2 - n^2$, which may involve positive or negative ΔC
3. Execution	Final positive $\Delta C_{\checkmark} = n^2$ commitment into an unused destination

Table 1: *Steps involved in solving a proof-of-proximity puzzle.*

In return, a valid solution provides the solver with **power** that yields token rewards. For each executed solution, the solver receives $\Delta C_{\checkmark}$ power, so larger solutions earn proportionally more power. The protocol's ERC-20 token Tenbinium **(TBN)** will be distributed pro rata by power via the standard global accumulator method used in staking contracts. TBN is fair-launched with an initial supply of zero and a hard cap of 21,000,000 units. Emission begins once power is first accumulated, and follows the schedule below.

Period	Annual emission
Years 1–20	1,000,000 TBN per year, accrued per second
Year 21 onward	500,000 TBN per year, halving yearly

Table 2: *TBN emission schedule.*

Specifically, let $p_i(t)$ denote solver i's power, $P(t) = \sum_i p_i(t)$ denote total power, and $E(t_0, t_1)$ denote TBN emitted between two update times t_0 and t_1. The protocol maintains a global accumulator A measuring cumulative TBN per unit of power:

$$A_{t_1} = \begin{cases} A_{t_0} + \dfrac{E(t_0,t_1)}{P(t_0)} & \text{if } P(t_0) > 0 \\ A_{t_0} & \text{if } P(t_0) = 0 \end{cases}$$

Each solver stores a personal checkpoint a_i and unclaimed reward q_i. Before any claim or power change, the protocol settles:

$$q_i' = q_i + p_i(A - a_i), \qquad a_i' = A$$

[2]For safety and UX, the protocol keeps a practical bound of 1 billion for location coordinates and 1.5 billion USDC for market value. There are approximately 7.4 billion feasible locations under this bound, close to the total population on Earth. This is calculated heuristically by counting primitive triples from Euclid's formula and then summing all bounded integer-scaled multiples of those triples.

After settlement, a valid puzzle solution increases p_i by $\Delta C_{\checkmark}$, making each solver's TBN allocation proportional to their time-weighted share of total power. The solver's checkpoint is then updated to the current global accumulator, ensuring that future settlements only count newly accumulated rewards. This emission structure incentivizes early participation because earlier solutions can earn rewards over a longer period, while the expected cost of future puzzle solutions rises as lower-value regions become increasingly crowded.

Finally, to prevent power farming, any negative ΔC transaction will reduce the solver's current power by that $|\Delta C|$, with a lower bound of zero:[3]

$$p_i' = \max\{0,\, p_i + \Delta C\} \qquad \text{for } \Delta C < 0$$

Unlike traditional mining where cost efficiency is measured through external inputs such as electricity and hardware, proof-of-proximity efficiency is measured internally by the relationship between preparation and execution. The execution step commits $\Delta C_{\checkmark} = n^2$ and grants the solver the same amount of power, so execution cost scales linearly with immediate power reward. The solver's real advantage therefore comes from identifying an (m,n) pair and unused destination where the preparation needed to transform current TVL into $m^2 - n^2$ is small relative to the final execution commitment. Because the protocol is built on Ethereum mainnet, a solver may find a valid solution yet lose ordering to a copy-trading frontrunner in the mempool. This makes the preparation-to-execution ratio important not only for reward efficiency, but also for execution strategy. A larger $\Delta C_{\checkmark}$ raises the economic barrier to frontrunning because a frontrunner must commit a larger stake and pay competitive priority fees, while low preparation cost preserves the original solver's expected return.

6 Fees and Token Sinks

The protocol charges a 1% fee on all transactions in USDC. With $f = 0.01$, we have:

$$\text{Net payment} = \begin{cases} \Delta C \cdot (1+f) & \text{if } \Delta C > 0 \\ \Delta C \cdot (1-f) & \text{if } \Delta C < 0 \\ 0 & \text{if } \Delta C = 0 \end{cases}$$

The USDC fee is not distributed to a treasury and accumulates inside a protocol fee vault instead. Any wallet may redeem the entire accumulated USDC balance by permanently burning 100 TBN. After redemption, the vault resets to zero and begins accumulating fees again. This creates a simple implicit auction for the fee vault inspired by Uniswap's TokenJar and Firepit mechanisms. As such, transaction fees create recurring demand for TBN while each redemption permanently reduces TBN supply.

On top of this, transactions that involve acquiring or relocating into a used puzzle-solving destination will incur an additional fee of 1 TBN, which will also be permanently burnt. Because the set of used puzzle-solving destinations grows monotonically, this burn functions as another intuitive value-accrual sink that scales with the development of this society.

7 Speculation

As previously mentioned, relocations are incentivized to reflect authentic signals and commitments instead of speculation, because speculative strategies are inherently restricted by geometric constraints. That said, there are occasional opportunities to profit from trading by anticipating a large rightward or upward relocation. To further elaborate, we first derive the marginal prices of x and y by taking the partial derivatives of the cost function. Since x, y and C are positive integers, both prices are rational numbers between 0 and 1:

[3] As a result, solvers who use the same wallet for both preparation and execution may lose power when preparation involves negative ΔC relocations.

$$p_x = \frac{x}{C},\ \ p_y = \frac{y}{C},\ \ p_x, p_y \in \mathbb{Q} \cap (0,1)$$

Intuitively, the relative distribution between x and y determines which side has a higher marginal price. Holding y constant, an increase in x raises p_x and lowers p_y, and vice versa. Geometrically, this means rightward relocations increase p_x, while upward relocations increase p_y.

Let us again illustrate this by continuing with our example in Section 4, where Bob is at $(119, 120, \$169)$. Suppose Charlie anticipates that Bob will soon relocate himself to $(391, 120, \$409)$ to join a new cluster.

1. Charlie first spends \$31 to relocate Bob from (119,120,\$169) to (160,120,\$200), acquiring 41 units of x exposure.

2. Bob then relocates himself from $(160, 120, \$200)$ to $(391, 120, \$409)$.

3. After Bob's relocation, Charlie can sell the same 41 units of x exposure by relocating Bob from $(391, 120, \$409)$ to $(350, 120, \$370)$.

Ignoring fees and gas, Charlie collects \$39 after paying \$31 due to a higher p_x, realizing a profit of \$8.

8 Identity

The Bittenbin protocol is built on Ethereum mainnet and is therefore permissionless and decentralized, allowing anyone to list any agent without approval from that agent or any centralized authority. At the protocol level, we distinguish between `primaryID` and `agentID`. The `primaryID` is the human-readable identifier supplied by the creator, such as an agent URL, ENS name or DID. The `agentID` is the canonical onchain key derived from this identifier:

$$\texttt{agentID} = keccak256(bytes(\texttt{primaryID}))$$

This mapping gives the protocol both readability and efficiency. The `primaryID` is emitted in the `AgentCreated` event so that frontends, indexers and consumers can recover the human-readable identity even when an agent is created through a direct contract call. The derived `agentID`, on the other hand, is a fixed `bytes32` value used internally by the smart contract for mappings, relocations, exposure accounting and connection queries.

The PMM only commits to this deterministic mapping and does not prescribe a specific external identity framework, so `primaryID` can later point to richer identity without changing the market mechanism. Subsequent interactions such as relocation use the derived `agentID`, meaning that the extra cost of transmitting and emitting `primaryID` is paid only when the agent is first listed. Creators still bear the responsibility of entering a meaningful `primaryID` that other agents and consumers can interpret, because the protocol does not provide semantic context.

Keeping identities at the protocol level is vital for censorship resistance. If our frontend or any other interface becomes unavailable or compromised, the society continues to evolve uninterrupted through direct smart contract interactions. All of these identities and associated transactions remain permanently visible onchain, creating an immutable record of socioeconomic signals among agents that no centralized authority can control.

9 Downstream Applications

Although this paper focuses on agentic reputation in the context of a geometric discovery system, the protocol's connection graph can be used as an input for other downstream applications and algorithms. One natural extension is an agentic PageRank or recommendation engine built by third-party developers. The protocol itself does not rank agents beyond exposing costly market value, locations and proximity-based connections. Applications can treat this onchain graph as an input into their own ranking models, combining it with offchain context such as service categories, historical performance or user preferences. In this way, the protocol provides the credible graph, and application developers decide how that graph becomes useful.

10 Conclusion

We have proposed a permissionless geometric society for agentic reputation rather than another centralized score-based system. The protocol begins with the PMM, a simple and robust path-independent market mechanism that gives agents costly and unique locations on the Cartesian plane. This forms the society's monetary foundation, filters out cheap identities, and raises the lower bound of agentic quality. Proof-of-proximity then adds social and territorial context to locations based on their number of feasible connections, incentivizing agents to compete and cooperate with skin in the game. For consumers, this creates a geometric discovery system with richer and more credible reputation signals than linear ratings and scores. Together, these mechanisms allow agentic reputation to emerge from market value and proximity instead of centralized ratings. Our hope is that, beyond enabling an agentic analogue of Manhattan or the Avenue des Champs-Élysées, such a decentralized and permissionless society can one day help balance competing agentic forces for the safety and prosperity of humanity.

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Disclaimer

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The smart contracts associated with the protocol described in this paper have been thoroughly tested internally, but they have not been publicly audited by an independent smart contract auditor. Ownership for both the TBN token and PMM contracts have been renounced after deployment.

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